



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: DSA0101

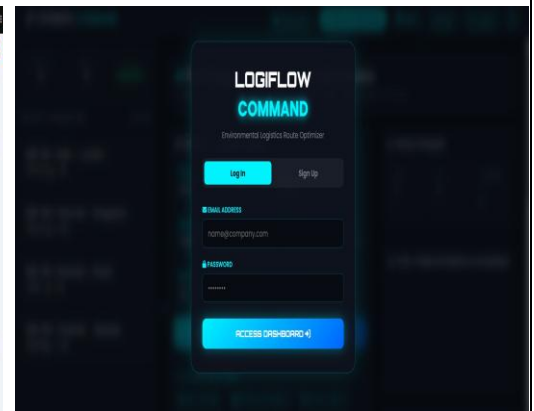
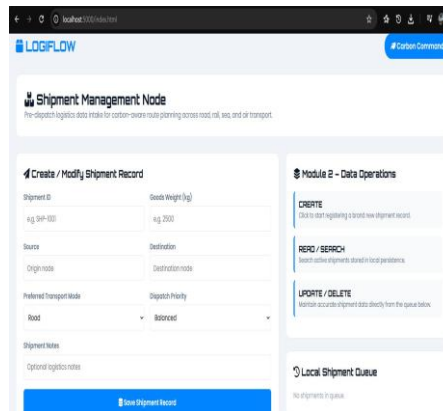
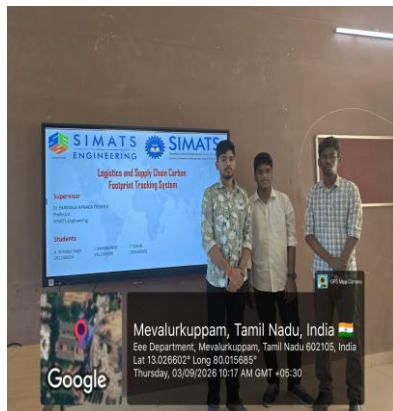
Slot: D

Course Name: OBJECT ORIENTED PROGRAMMING (OOP) IN C++

Course Faculty: Dr. Parimala Kanaga Devan K

Project Title: LOGISTICS AND SUPPLY CHAIN CARBON FOOTPRINT TRACKING SYSTEM

Module Photographs:



Project Description:

Module 1 serves as the core pre-transit routing engine for the C++ application, focusing directly on Path Optimization and Eco-Elasticity Routing. It calculates and optimizes freight shipment routes before dispatch to minimize carbon emissions while balancing overall transit times across multi-modal networks, including Road, Rail, Sea, and Air. At its center, the engine uses a lightweight, dual-weighted Dijkstra algorithm based on a combined cost formula: $\text{Cost} = \alpha \cdot \text{Time} + \beta \cdot \text{CO}_2$. This logic is structured through primary C++ Object-Oriented classes, including Graph for network topology management, PathOptimizer to compute eco-friendly routes, and an abstract TransportMode class with concrete subclasses (RoadTransport, RailTransport, SeaTransport, AirTransport) that encapsulate specific emission factors. Ultimately, Module 1 evaluates intermodal transfer hubs and delivers optimal green routes with sub-millisecond execution speeds.

Student Signature

Guide Signature